Advanced Cognitive Neuroscience

Week 43: Inverse modelling: Beamforming

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The course plan

Week 37:
Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:
Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:
Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:
Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:
Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)
Deadline for feedback: Video Explainer

Week 43:
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Wednesday: Inverse modelling: beamforming (Laura) (Coding)

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Wednesday: Work on your reports (Laura)

Week 45:
Monday: What about that other cortex? - the cerebellar one (Lau)
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Week 46:
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Wednesday: Respiration (Laura)
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Week 47:
No teaching: prepare presentations!

Week 48:
Monday: Oral presentations (part 1) (Lau)
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Week 49:
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Week 50:
Wednesday: What was it all about (Lau)

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Feedback on videos has now been published
For understanding of feedback (excellent, very good, good etc. refer to the grading scale:

https://international.au.dk/life/studentscomingtoau/student-life/studying-at-aarhus-university/exam-and-grading/

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TODAY

Beamforming

&

Creating a glossary of terms for the course so far

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Learning goals

- Getting an understanding of the beamformer;
  - how it reconstructs sources
  - how it dampens non-brain sources
  - how it can be applied across the whole brain
  - understanding the code for applying it
- Getting the dots connected so far

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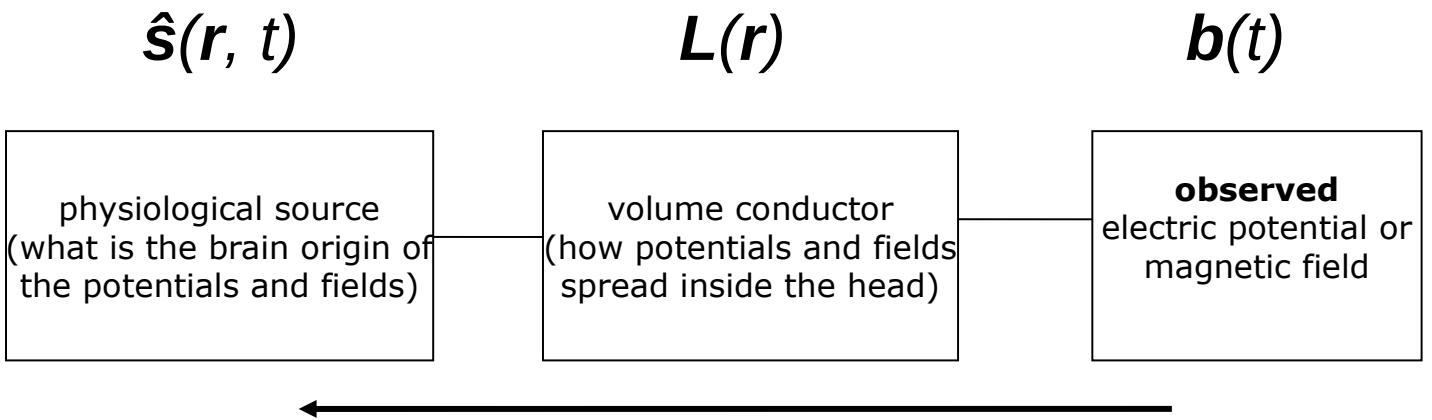
The problem in a nutshell

$$\boldsymbol{b}(t) = \boldsymbol{L}(r)\boldsymbol{s}(\boldsymbol{r}, t) + \boldsymbol{n}(t)$$

b*(t)*: the measured magnetic field at time *t*
s*(**r**, t)*: the sources at position ***r*** at time *t*
L*(r):* leadfield of sources a position ***r*** (a weighting matrix)
n(*t*): normally distributed noise at each time *t*

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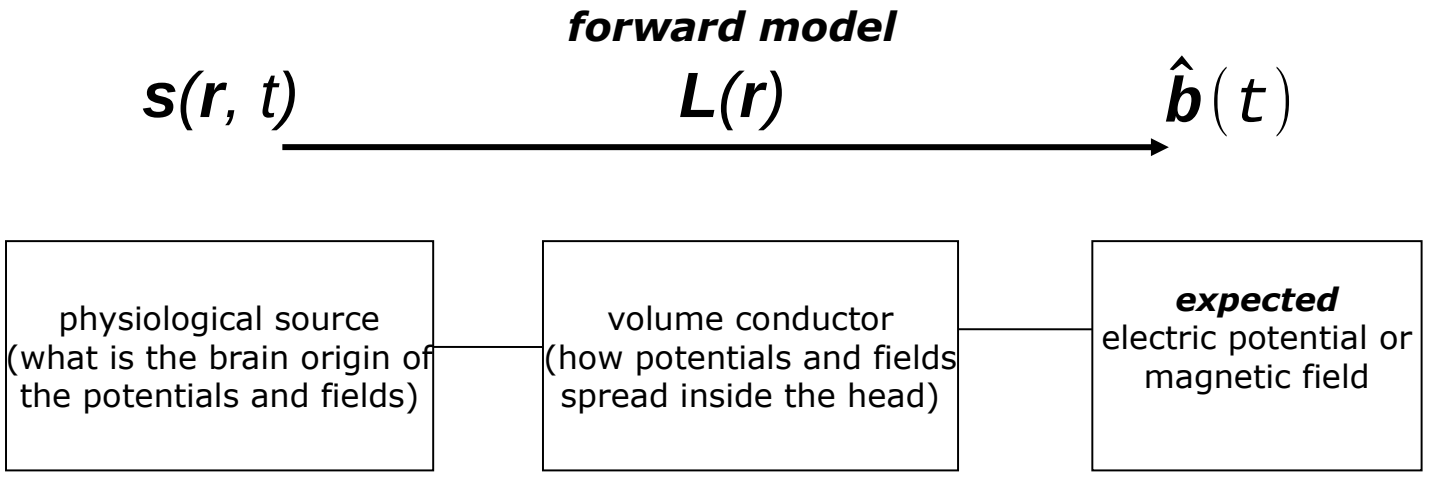
Inverse modelling



inverse model (source reconstruction)

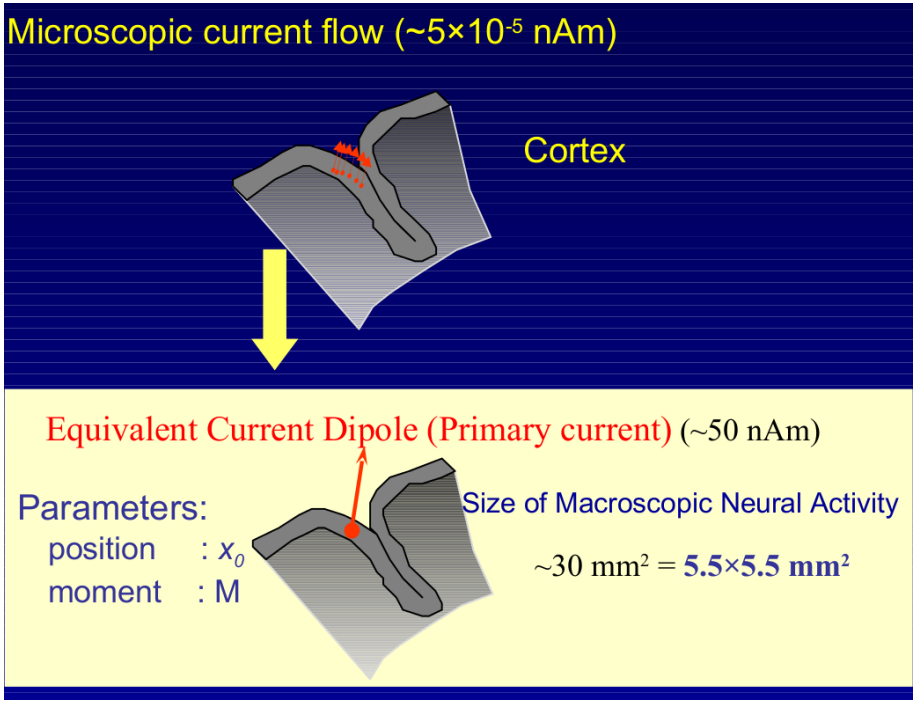
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Forward modelling



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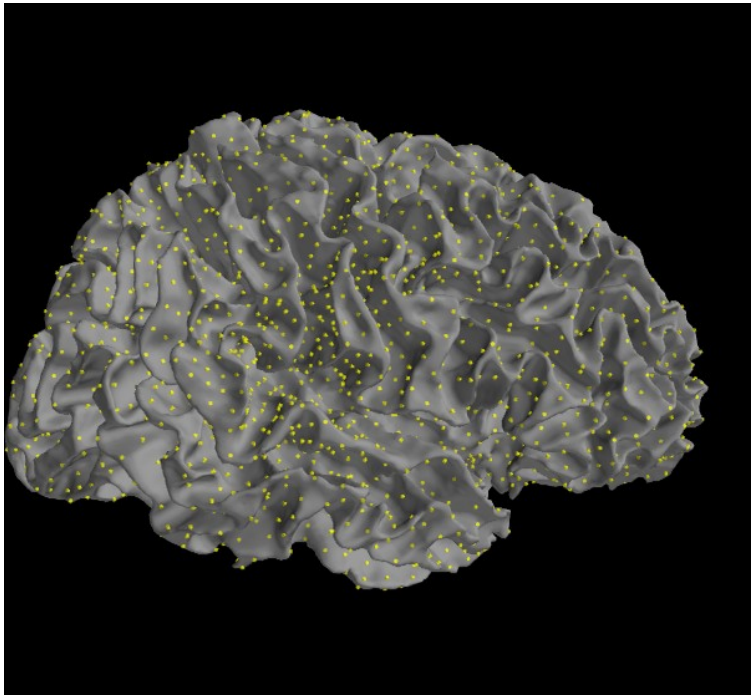
Equivalent Current Dipole



Stephanie Sillekens

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Source model



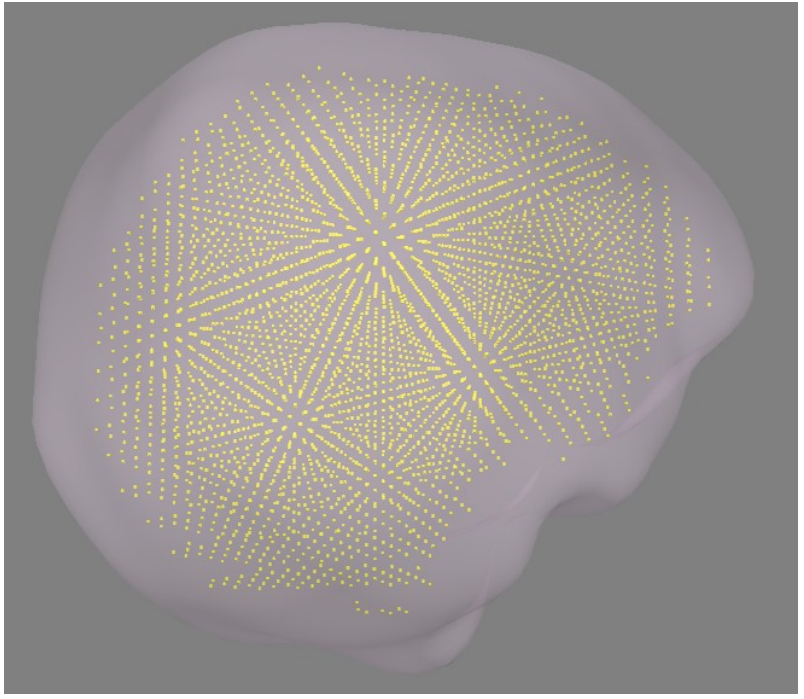
Restricted to the cortical surface

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https://github.com/ualsbombe/2025_advanced_cognitive_neuroscience/blob/main/experiment/mr_processing/python/bem_processing.py

Source model

VOLUMETRIC - UNRESTRICTED



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Inverse model types

- Dipole fitting
  - Only a single or a few dipolar sources are active at given time sample
  - The solution is *overdetermined:* there are more data channels (hundreds) than sources (a few)

$$[\boldsymbol{b}(t)-\boldsymbol{L}(\boldsymbol{r})\boldsymbol{L}(\boldsymbol{r})^{+}\boldsymbol{b}(t)]^{2}$$

- Minimum Norm Estimate
  - All (cortical) sources are active all the time
  - The solution is *underdetermined*: there are more sources (thousands) than channels (hundreds)

$$\hat{\boldsymbol{v}}_{\boldsymbol{vox}}(t)=\boldsymbol{L}_{\boldsymbol{V}}^{T}(\boldsymbol{G}+\epsilon\boldsymbol{I})^{(-1)}\boldsymbol{b}(t)$$

- Beamformer
  - All sources are estimated independently from one another

$$\boldsymbol{w}(\boldsymbol{r})=\frac{\boldsymbol{R}^{-1}\boldsymbol{L}(\boldsymbol{r})}{\boldsymbol{L}^{T}(\boldsymbol{r})\boldsymbol{R}^{-1}\boldsymbol{L}(\boldsymbol{r})}$$

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Beamformer

$$\hat{\boldsymbol{s}}(\boldsymbol{r}, t) = \boldsymbol{w}^{T}(\boldsymbol{r})\boldsymbol{b}(t)$$

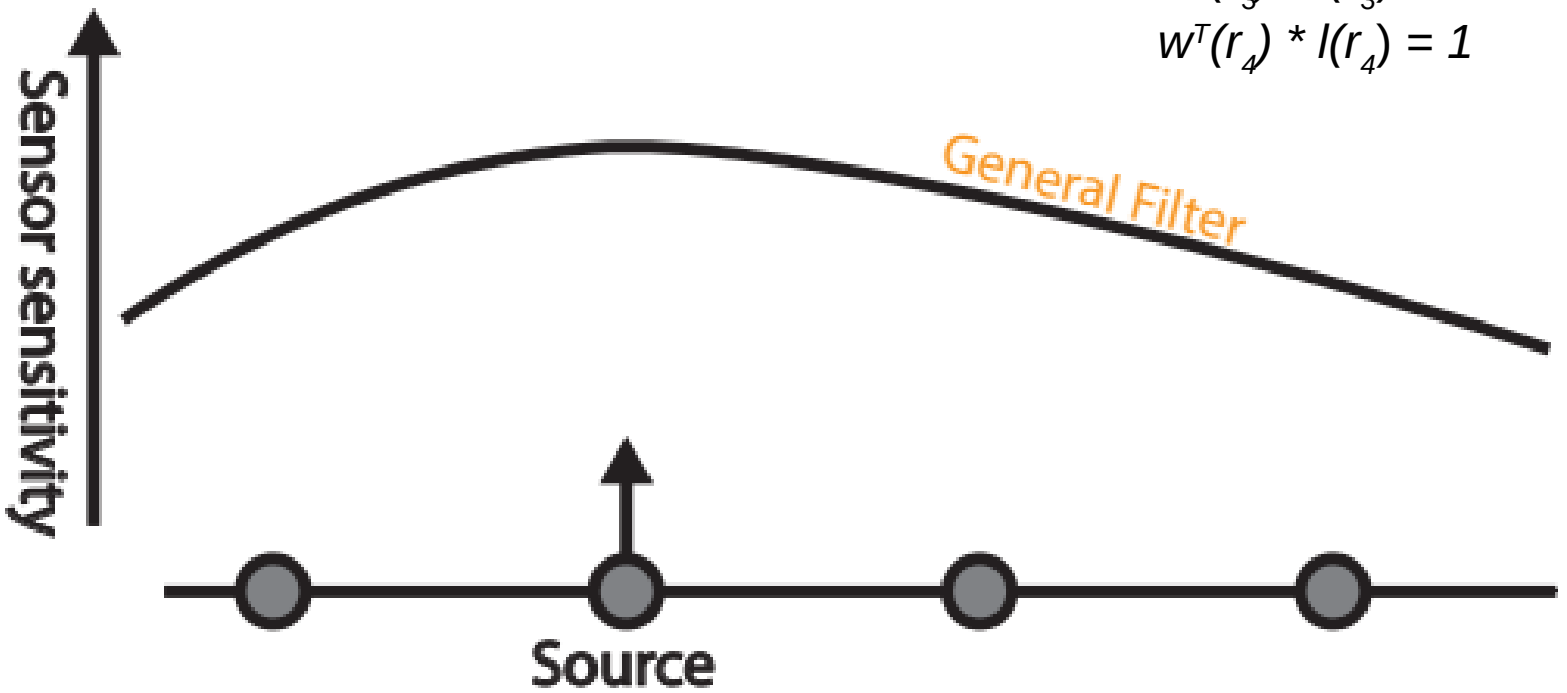
- For each source, ***r***, find a spatial filter, $\boldsymbol{w}^{T}$, that when multiplied onto the observed magnetic field or electric potential, ***b****(t)*, returns a good estimate of the underlying source activation, $\hat{s}(\boldsymbol{r}, t)$, for each time point, *t*

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Spatial sensitivity and leakage of a filter

Unity passband

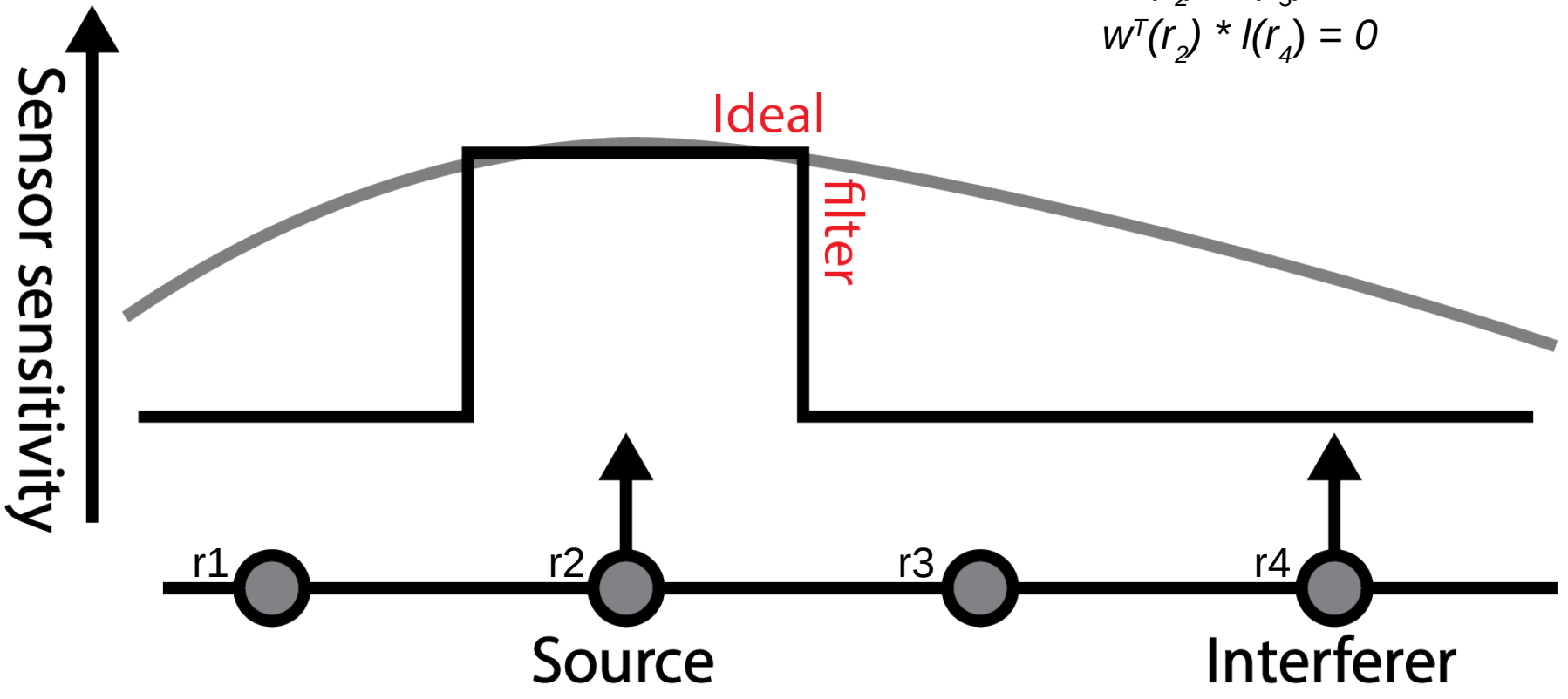
$$w^T(r_1) * l(r_1) = 1$$
$$w^T(r_2) * l(r_2) = 1$$
$$w^T(r_3) * l(r_3) = 1$$
$$w^T(r_4) * l(r_4) = 1$$



Spatial sensitivity and leakage of a filter

Unity passband
Minimum beamformer output

$$w^T(r_2) * l(r_1) = 0$$
$$w^T(r_2) * l(r_2) = 1$$
$$w^T(r_2) * l(r_3) = 0$$
$$w^T(r_2) * l(r_4) = 0$$

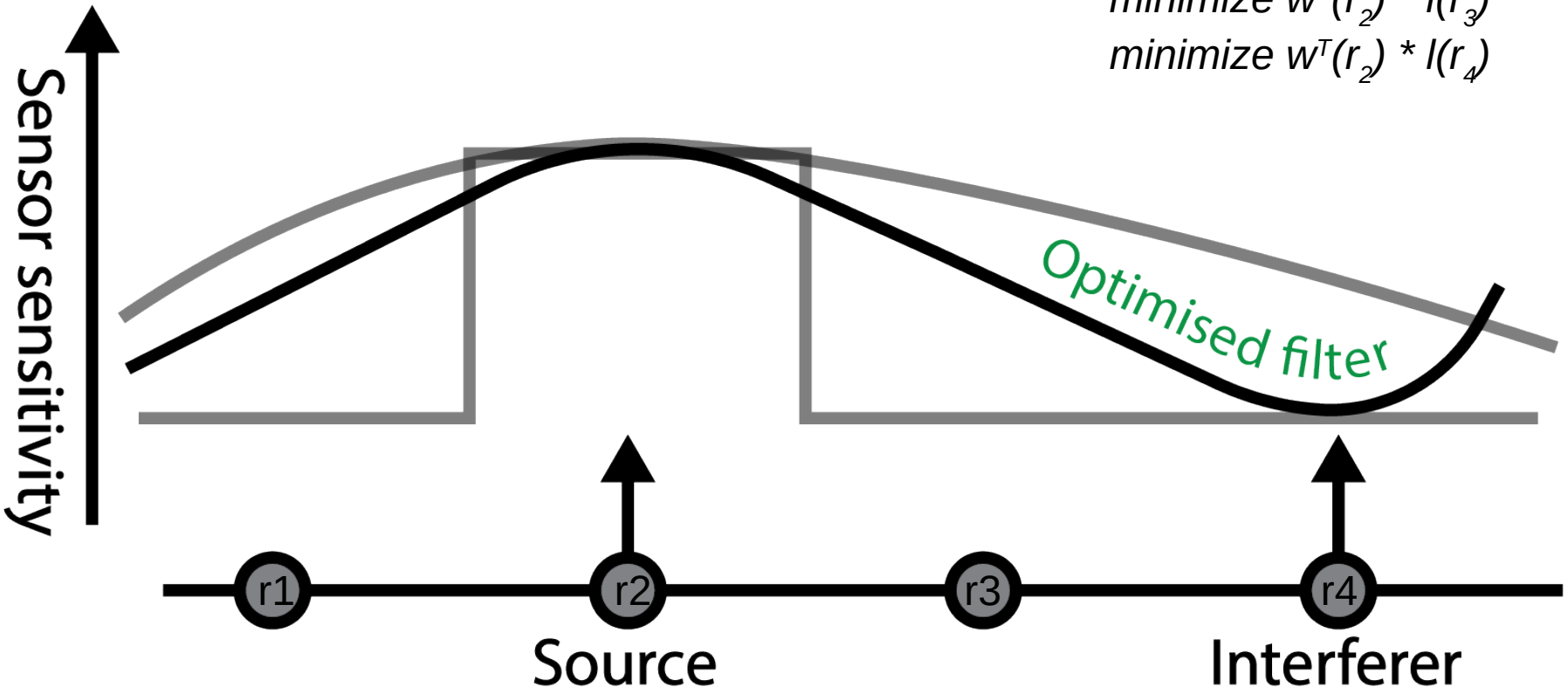


Spatial sensitivity and leakage of a filter

Unity passband

Minimum beamformer output

$minimize\ w^T(r_2) * l(r_1)$
$w^T(r_2) * l(r_2) = 1$
$minimize\ w^T(r_2) * l(r_3)$
$minimize\ w^T(r_2) * l(r_4)$



Beamformer

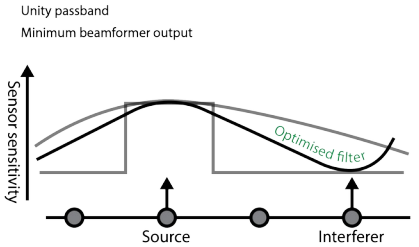
$$\boldsymbol{w}(\boldsymbol{r}) = \underset{\boldsymbol{w}(\boldsymbol{r})}{\operatorname{argmin}}\ \boldsymbol{w}^{\boldsymbol{T}}(\boldsymbol{r})\boldsymbol{R}\boldsymbol{w}(\boldsymbol{r}), \text{ subject to } \boldsymbol{w}^{\boldsymbol{T}}(\boldsymbol{r})\boldsymbol{L}(\boldsymbol{r}) = 1$$

Minimization of variance: $\underset{\boldsymbol{w}(\boldsymbol{r})}{\operatorname{argmin}}\ \boldsymbol{w}^{\boldsymbol{T}}(\boldsymbol{r})\boldsymbol{R}\boldsymbol{w}(\boldsymbol{r})$

Unit-gain constraint: $\boldsymbol{w}^{\boldsymbol{T}}(\boldsymbol{r})\boldsymbol{L}(\boldsymbol{r}) = 1$

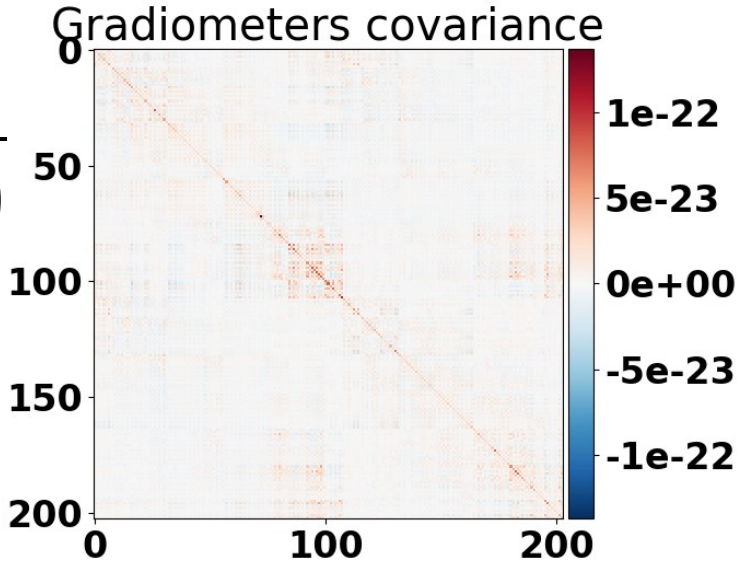
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Beamformer



Data covariance matrix (*R*)

$$w(r) = \frac{R^{-1}L(r)}{L^{T}(r)R^{-1}L(r)}$$



***w(r)* =** spatial filter
L(r) = lead field or forward model
R = data covariance matrix

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I’ll now derive this

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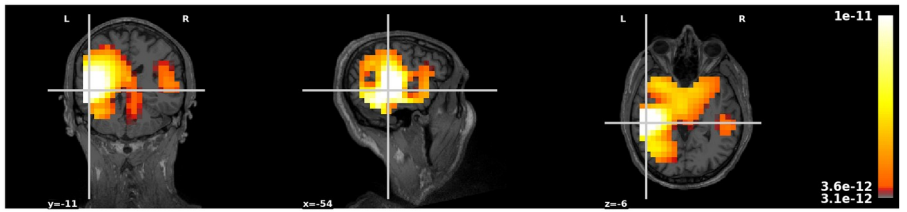
Lagrange

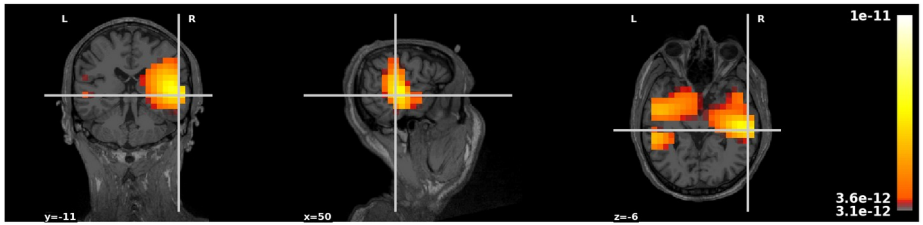
https://www.youtube.com/watch?v=hQ4UNu1P2kw

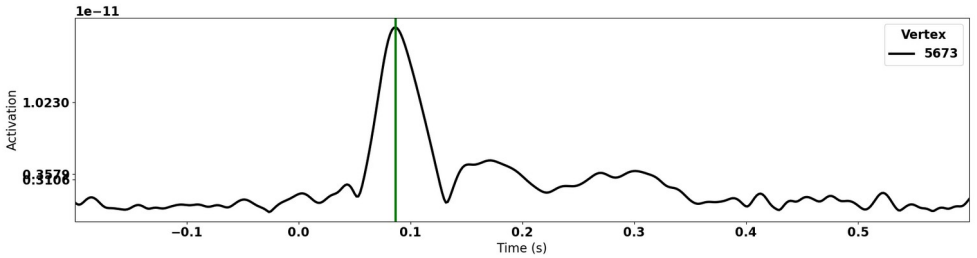
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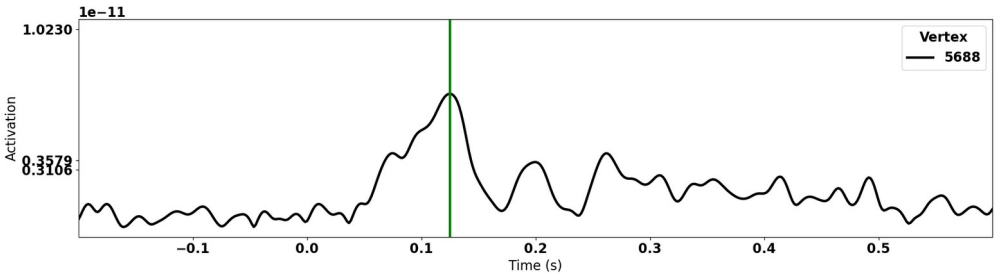
86 ms

125 ms











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CODING EXAMPLE
beamformer w43

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Summary and different strategies

- Dipole fitting and minimum norm estimate:
  - optimize fit between model and data, i.e. there is a relation between the brain map and the sensor data
- Beamformer
  - Estimate each source independently by minimizing the influence of other sources
    - Thus very good at removing the influence from non-brain sources, such as eye-blinks and the like (because they are not associated with a leadfield from the brain)
  - Thus really good for estimating time courses of sources chosen *a priori,* but the relation between the brain map and the sensor data is less clear than for the minimum norm estimate

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Glossary of terms

- Super-terms to create glossary around
  - Basic physiology
  - Multivariate statistics
  - Forward modelling (and dipole estimation)
  - Inverse modelling; MNE and beamforming

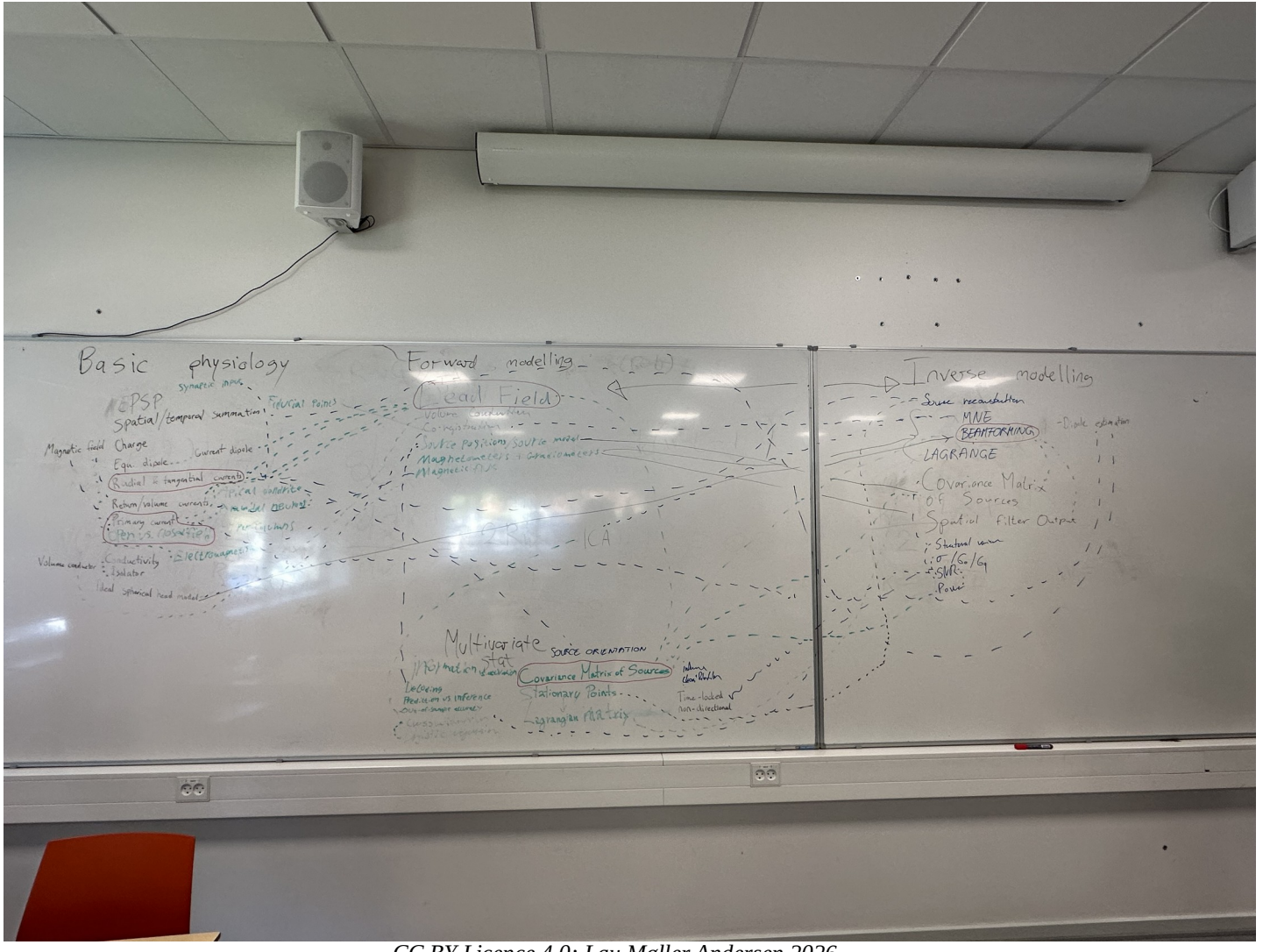
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Glossary of terms

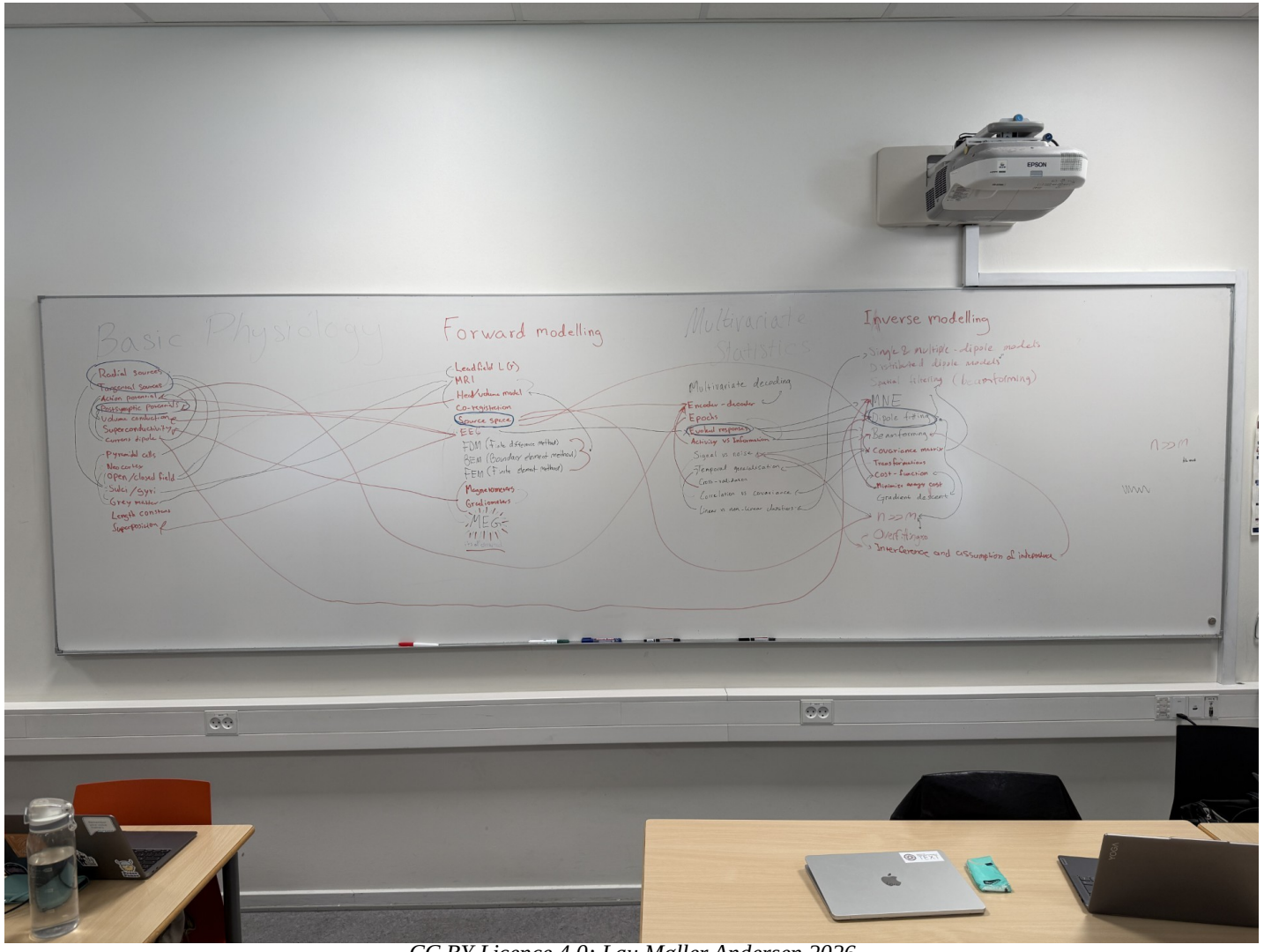
PROCEDURE

1) (10 minutes): in your study groups find all the sub-terms you think belong to each of the four super-terms
2) (10 minutes) in pairs of groups, go and write the sub-terms for each term on the whiteboard; start making internal arrows connecting them
3) (10 minutes) start connecting the sub-terms and the super-terms all across the board (mayhem will ensue)
4) (10 minutes) discussing it all together
5) Take pictures to preserve for posterity

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Reading questions for W45

- Sokolov et al. 2017
  - What is a forward model in sensorimotor control?
  - What does it mean for predictions being cognitive?
  - What is error-based learning
- Andersen and Dalal 2024
  - What are the proposed different roles of basal ganglia and cerebellum in relation to timing?
  - What is proactive action?
  - What would proactive cognition be? (Think back to Sokolov)

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Derivation from the whiteboard

$$\mathscr{L}(\boldsymbol{w}, \kappa) = \boldsymbol{w}^T \boldsymbol{R} \boldsymbol{w} + \kappa(\boldsymbol{w}^T \boldsymbol{L}(\boldsymbol{r}) - 1) \quad \text{(eq 1)}$$

$$\frac{\delta \mathscr{L}(\boldsymbol{w}, \kappa)}{\delta \boldsymbol{w}} = 2\boldsymbol{R}\boldsymbol{w} + \kappa \boldsymbol{L}(\boldsymbol{r}) \quad \text{(eq 2)}$$

Finding the minimum

$$2\boldsymbol{R}\boldsymbol{w} + \kappa \boldsymbol{L}(\boldsymbol{r}) = 0 \quad \text{(eq 3)}$$

$$\boldsymbol{w} = -\kappa \boldsymbol{R}^{-1} \boldsymbol{L}(\boldsymbol{r})/2 \quad \text{(eq 4)}$$

From substituting $\boldsymbol{w}$ into: $\boldsymbol{w}^T \boldsymbol{L}(\boldsymbol{r}) = 1$

follows: $\kappa = \frac{-2}{\boldsymbol{L}^T(\boldsymbol{r}) \boldsymbol{R}^{-1} \boldsymbol{L}(\boldsymbol{r})}$ (eq 5)

Then subtistuting κ into eq 4 we get:

$$\boldsymbol{w}(\boldsymbol{r}) = \frac{\boldsymbol{R}^{-1} \boldsymbol{L}(\boldsymbol{r})}{\boldsymbol{L}^T(\boldsymbol{r}) \boldsymbol{R}^{-1} \boldsymbol{L}(\boldsymbol{r})}$$

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